

UNIVERSITY OF MALAYA

EXAMINATION FOR THE DEGREE OF MASTER OF DATA SCIENCE

ACADEMIC SESSION 2019/2020 : SEMESTER I

WQD7006 : Machine Learning for Data Science

Dec 2019/Jan 2020

Time : 3 hours

INSTRUCTIONS TO CANDIDATES :

Answer **ALL** questions (100 marks).

(This question paper consists of 5 questions on 6 printed pages)

1. Answer the followings with regard to machine learning for data science. Include suitable diagrams where appropriate to assist your explanation. Every part below carries 2 marks each.

- (a) What are the major differences between supervised learning and unsupervised learning?
- (b) Differentiate between generative and discriminative classifiers.
- (c) How does learning rate of Gradient Descent algorithm affect its outcome with regard to local minimum and global minimum?
- (d) Sketch a diagram to show a Multi-Layer Perceptron (MLP) with one hidden layer where each of them has 3 neurons. Label every part of the MLP clearly.
- (e) List the steps of a sequential back-propagation training.
- (f) Explain interactions among components of a Reinforcement Learning model.
- (g) Describe the procedures of a 5-fold cross validation.
- (h) Identify 4 advantages of Naïve Bayes classifiers.
- (i) How does the value of k affect the outcome of a k -Nearest Neighbor classifier?
- (j) Determine 3 principal components in a Principal Component Analysis of face recognition. Then discuss how they can offer dimensionality reduction for face recognition.

(20 marks)

[20 marks]

2. Consider a dataset that describes the weather conditions for playing a lawnball game in Table 2.1. Given the weather conditions, each tuple classifies the conditions as fit(Yes) or unfit(No) for playing lawnball.

	Outlook	Temperature	Humidity	Windy	Play lawnball
1	Rainy	Hot	High	False	No
2	Rainy	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Sunny	Mild	High	False	Yes
5	Sunny	Cool	Normal	False	Yes
6	Sunny	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Rainy	Mild	High	False	No
9	Rainy	Cool	Normal	False	Yes
10	Sunny	Mild	Normal	False	Yes
11	Rainy	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Sunny	Mild	High	True	No

Table 2.1 Weather Conditions for Playing Lawnball

	Yes	No	P(Yes)	P(No)
Sunny				
Overcast				
Rainy				
Total	9	5	100%	100%

Table 2.2 Probability of Playing Lawnball according to Weather Outlook

Play Lawnball	P(Yes)/P(No)
Yes	
No	
Total	100%

Table 2.3 Class Probabilities

Answer the following parts:

- Write an equation to describe Bayes' Theorem. (2 marks)
- Fill up Table 2.2 by computing the probability for each case of Yes and No according to the dataset in Table 2.1 (4 marks)
- Repeat the task in (b) above for other features of Temperature, Humidity, and Windy. (12 marks)
- Fill up Table 2.3. (2 marks)

- (e) Compute the probability of playing Lawnball today $P(\text{Yes}|\text{today})$ using the results above if the weather today is Sunny, Hot, Normal Humidity, and Not Windy.

(2 marks)

- (f) Repeat the computation in (e) above for the probability of not playing Lawnball today $P(\text{No}|\text{today})$.

(2 marks)

- (g) Conclude the prediction result for playing Lawnball today using your answers in (e) and (f) above.

(1 mark)

[25 marks]

3. Classification is part of supervised learning.

- (a) Give 2 example applications of binary classification.

(2 marks)

- (b) Identify 2 applications of multi-class classification.

(2 marks)

- (c) Is Logistic Regression suitable for weather forecasting? Justify your answer.

(2 marks)

- (d) Illustrate support vectors in Support Vector Machines (SVM). Then, use the diagram to explain SVM's optimization objective.

(2 marks)

- (e) Maximum depth and number of samples in tree nodes are hyper-parameters in Decision Tree. How do they cause over-fitting and under-fitting in Decision Tree?

(4 marks)

- (f) How does Random Forest overcome overfitting of a Decision Tree?

(2 marks)

- (g) List the steps of an ensemble learning algorithm. Provide a suitable sketch to relate this algorithm to the Random Forest method.

(8 marks)

- (h) What parameter needs tuning in the Random Forest method? How does this parameter affect the performance of the Random Forest method?

(3 marks)

[25 marks]

4. Given a dataset below.

x	y	z
1	2	3
2	4	6
3	6	9
4	8	12
5	10	15
6	12	18

(a) Design a suitable method to compress it by 50% assuming each number takes one byte of storage space.

(5 marks)

(b) Data compression is similar with dimensionality reduction. Name one dimensionality reduction method. Then write its algorithm in pseudocodes.

(5 marks)

(c) Identify 2 applications of dimensionality reduction method.

(2 marks)

(d) Feature extraction can be applied in dimensionality reduction. How do eigenvectors and eigenvalues of a multi-dimensional dataset indicate principal component of features extracted?

(3 marks)

[15 marks]

5. Artificial neuron is the basic unit of an Artificial Neural Network (ANN).

(a) Draw an artificial neuron and describe the functions of its components.
(5 marks)

(b) Compare the structure of a biological neuron with the structure of an artificial neuron using suitable diagrams.
(6 marks)

(c) Compare the advantages and disadvantages between supervised learning and unsupervised learning in ANN.
(4 marks)

[15 marks]

END